Class 9 Mathematics — Exercise 4.1

Basic Concept

(1) Factorization

The process in which an algebraic expression is written as the **product of its factors** is called factorization.

$$15x^2y = (5x)(3xy)$$

Here $5x$ and $3xy$ are the factors of $15x^2y$. Factorization is the **opposite of multiplication**:

$$\text{Factors } (5x)(3xy) \xrightarrow{\text{Multiply}} 15x^2y \qquad 15x^2y \xrightarrow{\text{Factorize}} (5x)(3xy)$$

(2) Formulas Used

	Type	Factors
(i)	$ka + kb + kc$	$k(a + b + c)$
(ii)	$ac + ad + bc + bd$	$(a + b)(c + d)$
(iii)	$a^2 + 2ab + b^2$	$(a + b)^2$
(iv)	$a^2 - 2ab + b^2$	$(a - b)^2$
(v)	$a^2 - b^2$	$(a + b)(a - b)$
(vi)	$a^2 + 2ab + b^2 - c^2$	$(a + b + c)(a + b - c)$
(vii)	$a^2 - 2ab + b^2 - c^2$	$(a - b + c)(a - b - c)$

(3) Splitting the Middle Term ($ax^2 + bx + c$)

Multiply the first and last coefficients ($a \times c$). Find two numbers whose **product** is $a \times c$ and whose **sum** is b. Split the middle term using these two numbers, then take common.

Question 1

$$2x^2y^3 - 6x^2y^2 + 2xy^3$$

$$= \boxed{2xy^2(xy - 3x + y)} \text{ Ans}$$

Question 2

$$3nx - 3x - 3ny + 3y$$

$$= 3[nx - x - ny + y]$$
$$= 3[x(n-1) - y(n-1)]$$
$$= \boxed{3(n-1)(x-y)} \text{ Ans}$$

Question 3

$$18x^4 + 108x^2y^2 + 162y^4$$

$$= 18[x^4 + 6x^2y^2 + 9y^4] \qquad (\because a^2 + 2ab + b^2 = (a+b)^2)$$
$$= 18[(x^2)^2 + 2(x^2)(3y^2) + (3y^2)^2]$$
$$= \boxed{18(x^2 + 3y^2)^2} \text{ Ans}$$

Question 4

$$(k+2)^2 - 8(k+2) + 16$$

$$= (k+2)^2 - 2(k+2)(4) + (4)^2 \qquad (\because a^2 - 2ab + b^2 = (a-b)^2)$$
$$= (k+2-4)^2$$
$$= \boxed{(k-2)^2} \text{ Ans}$$

Question 5

$$9x^2 + 4 - 169y^2 - 12x$$

$$= 9x^2 + 4 - 12x - 169y^2$$
$$= [(3x)^2 + (2)^2 - 2(3x)(2)] - (13y)^2 \qquad (\because a^2 + b^2 - 2ab = (a-b)^2)$$
$$= (3x-2)^2 - (13y)^2 \qquad (\because a^2 - b^2 = (a+b)(a-b))$$
$$= (3x - 2 + 13y)(3x - 2 - 13y)$$
$$= \boxed{(3x + 13y - 2)(3x - 13y - 2)} \text{ Ans}$$

Question 6

$$(x^2-1)(y+1) - (y+3)(x^2-1)$$

$$= (x^2-1)[(y+1) - (y+3)]$$
$$= (x^2-1)[y + 1 - y - 3]$$
$$= (x^2-1)(1-3)$$
$$= (x^2-1)(-2)$$
$$= -2(x^2 - 1^2) \qquad (\because a^2 - b^2 = (a+b)(a-b))$$
$$= \boxed{-2(x+1)(x-1)} \text{ Ans}$$

Question 7

$$x^2 - 6ax + 9a^2 - 16b^2$$

$$= [(x)^2 - 2(x)(3a) + (3a)^2] - (4b)^2$$
$$= (x - 3a)^2 - (4b)^2 \qquad (\because a^2 - b^2 = (a + b)(a - b))$$
$$= \boxed{(x - 3a + 4b)(x - 3a - 4b)} \text{ Ans}$$

Question 8

$$1 - x^2 - 2xy - y^2$$

$$= 1 - (x^2 + 2xy + y^2)$$
$$= (1)^2 - (x + y)^2 \qquad (\because a^2 - b^2 = (a + b)(a - b))$$
$$= [1 + (x + y)][1 - (x + y)]$$
$$= \boxed{(1 + x + y)(1 - x - y)} \text{ Ans}$$

Question 9

Find a polynomial whose factorization is $(x+y-2c)(x+2c+y)$ by using an appropriate formula.

$$(x + y - 2c)(x + 2c + y)$$
$$= (x + y - 2c)(x + y + 2c) \qquad (\because (a - b)(a + b) = a^2 - b^2)$$
$$= (x + y)^2 - (2c)^2$$
$$= \boxed{x^2 + y^2 + 2xy - 4c^2} \text{ Ans}$$

Question 10

Show the expression $x^2 + 4y^2 - z^2 + 4xy$ as the difference of two squares.

$$x^2 + 4y^2 - z^2 + 4xy$$
$$= x^2 + 4y^2 + 4xy - z^2$$
$$= [(x)^2 + (2y)^2 + 2(x)(2y)] - (z)^2$$
$$= \boxed{(x + 2y)^2 - (z)^2} \text{ Ans}$$

Question 11

Find the missing factor in the following.

(a) $(2y^2 - 3y - 27) = (y + 3)(\quad)$

$2 \times (-27) = -54 \;\rightarrow\; -9$ and $+6 \quad (-9 + 6 = -3)$

$$\begin{aligned} & 2y^2 - 3y - 27 \\ &= 2y^2 + 6y - 9y - 27 \\ &= 2y(y + 3) - 9(y + 3) \\ &= (y + 3)(2y - 9) \end{aligned}$$

$\boxed{(2y - 9) \text{ is the missing factor}}$

(b) $(5x^2 + 12x - 9) = (\quad)(x + 3)$

$5 \times (-9) = -45 \;\rightarrow\; 15$ and $-3 \quad (15 - 3 = 12)$

$$\begin{aligned} & 5x^2 + 12x - 9 \\ &= 5x^2 + 15x - 3x - 9 \\ &= 5x(x + 3) - 3(x + 3) \\ &= (5x - 3)(x + 3) \end{aligned}$$

$\boxed{(5x - 3) \text{ is the missing factor}}$

Question 12

$$x^4 + 4m^4$$

$$\begin{aligned} &= (x^2)^2 + (2m^2)^2 + 2(x^2)(2m^2) - 2(x^2)(2m^2) \\ &= (x^2 + 2m^2)^2 - 4x^2m^2 \\ &= (x^2 + 2m^2)^2 - (2xm)^2 \qquad (\because a^2 - b^2 = (a + b)(a - b)) \\ &= \boxed{(x^2 + 2m^2 + 2xm)(x^2 + 2m^2 - 2xm)} \text{ Ans} \end{aligned}$$

Question 13

$$m^4 + m^2 + 1$$

$$\begin{aligned} &= m^4 + 1 + m^2 \\ &= (m^2)^2 + (1)^2 + 2(m^2)(1) - 2(m^2)(1) + m^2 \\ &= (m^2 + 1)^2 - 2m^2 + m^2 \\ &= (m^2 + 1)^2 - (m)^2 \qquad (\because a^2 - b^2 = (a + b)(a - b)) \\ &= \boxed{(m^2 + 1 + m)(m^2 + 1 - m)} \text{ Ans} \end{aligned}$$

Question 14

$$-3x^4 - 21x^3 + 24x^2$$

$$= -3x^2[x^2 + 7x - 8]$$

$-8 \rightarrow 8$ and $-1 \quad (8 - 1 = 7)$

$$= -3x^2[x^2 - x + 8x - 8]$$
$$= -3x^2[x(x-1) + 8(x-1)]$$
$$= \boxed{-3x^2(x-1)(x+8)} \text{ Ans}$$

Question 15

$$x^8 + x^4 + 1$$

$$= x^8 + 1 + x^4$$
$$= (x^4)^2 + (1)^2 + 2(x^4)(1) - 2(x^4)(1) + x^4$$
$$= (x^4 + 1)^2 - 2x^4 + x^4$$
$$= (x^4 + 1)^2 - x^4$$
$$= (x^4 + 1)^2 - (x^2)^2 \qquad (\because a^2 - b^2 = (a+b)(a-b))$$
$$= (x^4 + 1 + x^2)(x^4 + 1 - x^2)$$
$$= [(x^2)^2 + (1)^2 + 2(x^2)(1) - 2(x^2)(1) + x^2](x^4 + 1 - x^2)$$
$$= [(x^2 + 1)^2 - 2x^2 + x^2](x^4 + 1 - x^2)$$
$$= [(x^2 + 1)^2 - (x)^2](x^4 + 1 - x^2)$$
$$= \boxed{(x^2 + 1 + x)(x^2 + 1 - x)(x^4 + 1 - x^2)} \text{ Ans}$$

Question 16

$$4x^4 + 256y^4$$

$$= 4(x^4 + 64y^4)$$
$$= 4[(x^2)^2 + (8y^2)^2 + 2(x^2)(8y^2) - 2(x^2)(8y^2)]$$
$$= 4[(x^2 + 8y^2)^2 - 16x^2y^2]$$
$$= 4[(x^2 + 8y^2)^2 - (4xy)^2] \qquad (\because a^2 - b^2 = (a+b)(a-b))$$
$$= \boxed{4(x^2 + 8y^2 + 4xy)(x^2 + 8y^2 - 4xy)} \text{ Ans}$$

Question 17

$$12 - 7x + x^2$$

$$= x^2 - 7x + 12$$

$1 \times 12 = 12 \ \rightarrow \ -4$ and $-3 \quad (-4 - 3 = -7)$

$$= x^2 - 4x - 3x + 12$$
$$= x(x-4) - 3(x-4)$$
$$= \boxed{(x-4)(x-3)} \text{ Ans}$$

Question 18

$$x^2 - 9x + 8$$

$8 \ \rightarrow \ -8$ and $-1 \quad (-8 - 1 = -9)$

$$= x^2 - x - 8x + 8$$
$$= x(x-1) - 8(x-1)$$
$$= \boxed{(x-1)(x-8)} \text{ Ans}$$

Question 19

$$10z^2 - 29z + 10$$

$10 \times 10 = 100 \ \rightarrow \ -25$ and $-4 \quad (-25 - 4 = -29)$

$$= 10z^2 - 25z - 4z + 10$$
$$= 5z(2z-5) - 2(2z-5)$$
$$= \boxed{(2z-5)(5z-2)} \text{ Ans}$$

Question 20

$$-3y^2 + 13y - 4$$

$$= -[3y^2 - 13y + 4]$$

$3 \times 4 = 12 \ \rightarrow \ -12$ and $-1 \quad (-12 - 1 = -13)$

$$= -[3y^2 - 12y - y + 4]$$
$$= -[3y(y-4) - 1(y-4)]$$
$$= -(y-4)(3y-1)$$
$$= (-y+4)(3y-1)$$
$$= (4-y)(3y-1)$$
$$= \boxed{(3y-1)(4-y)} \text{ Ans}$$

Question 21

$$x^2 - 21x + 90$$

$90 \rightarrow -15 \text{ and } -6 \quad (-15 - 6 = -21)$

$$\begin{aligned} &= x^2 - 6x - 15x + 90 \\ &= x(x-6) - 15(x-6) \\ &= \boxed{(x-6)(x-15)} \text{ Ans} \end{aligned}$$

Question 22

$$x^2 + x - 2$$

$-2 \rightarrow 2 \text{ and } -1 \quad (2 - 1 = 1)$

$$\begin{aligned} &= x^2 + 2x - x - 2 \\ &= x(x+2) - 1(x+2) \\ &= \boxed{(x+2)(x-1)} \text{ Ans} \end{aligned}$$

Question 23

$$3x^2 + 11x + 6$$

$3 \times 6 = 18 \rightarrow 9 \text{ and } 2 \quad (9 + 2 = 11)$

$$\begin{aligned} &= 3x^2 + 9x + 2x + 6 \\ &= 3x(x+3) + 2(x+3) \\ &= \boxed{(x+3)(3x+2)} \text{ Ans} \end{aligned}$$

Question 24

$$2x^2 - 5xy - 3y^2$$

$2 \times (-3) = -6 \rightarrow -6 \text{ and } 1 \quad (-6 + 1 = -5)$

$$\begin{aligned} &= 2x^2 - 6xy + xy - 3y^2 \\ &= 2x(x-3y) + y(x-3y) \\ &= \boxed{(2x+y)(x-3y)} \text{ Ans} \end{aligned}$$

Question 25

$$8 + 6x - 5x^2$$

$8 \times (-5) = -40 \rightarrow 10 \text{ and } -4 \quad (10 - 4 = 6)$

$$\begin{aligned} &= 8 - 4x + 10x - 5x^2 \\ &= 4(2-x) + 5x(2-x) \\ &= \boxed{(2-x)(4+5x)} \text{ Ans} \end{aligned}$$

Question 26

$$6 - 7x - 5x^2$$

$6 \times (-5) = -30 \;\rightarrow\; -10 \text{ and } +3 \quad (-10 + 3 = -7)$

1st Method:

$$\begin{aligned} &= 6 + 3x - 10x - 5x^2 \\ &= 3(2 + x) - 5x(2 + x) \\ &= \boxed{(2 + x)(3 - 5x)} \text{ Ans} \end{aligned}$$

2nd Method:

$$\begin{aligned} 6 - 7x - 5x^2 &= -5x^2 - 7x + 6 \\ &= -5x^2 - 10x + 3x + 6 \\ &= -5x(x + 2) + 3(x + 2) \\ &= (x + 2)(-5x + 3) \\ &= (x + 2)(3 - 5x) \end{aligned}$$

Question 27

$$2a^2 - 4a - 6$$

$2 \times (-6) = -12 \;\rightarrow\; -6 \text{ and } +2 \quad (-6 + 2 = -4)$

1st Method:

$$\begin{aligned} &= 2a^2 + 2a - 6a - 6 \\ &= 2a(a + 1) - 6(a + 1) \\ &= (2a - 6)(a + 1) \\ &= 2(a - 3)(a + 1) \\ &= \boxed{2(a + 1)(a - 3)} \text{ Ans} \end{aligned}$$

2nd Method:

$$\begin{aligned} 2a^2 - 4a - 6 &= 2(a^2 - 2a - 3) \\ &= 2(a^2 - 3a + a - 3) \\ &= 2[a(a - 3) + 1(a - 3)] \\ &= 2(a - 3)(a + 1) \end{aligned}$$

Question 28

$$u^4 - 13u^2 + 36$$

$36 \rightarrow -9$ and $-4 \quad (-9 - 4 = -13)$

$$\begin{aligned} &= u^4 - 9u^2 - 4u^2 + 36 \\ &= u^2(u^2 - 9) - 4(u^2 - 9) \\ &= (u^2 - 9)(u^2 - 4) \\ &= [(u)^2 - (3)^2][(u)^2 - (2)^2] \\ &= \boxed{(u+3)(u-3)(u+2)(u-2)} \text{ Ans} \end{aligned}$$

Question 29

$$y^4 - 12y^2 - 64$$

$-64 \rightarrow -16$ and $+4 \quad (-16 + 4 = -12)$

$$\begin{aligned} &= y^4 - 16y^2 + 4y^2 - 64 \\ &= y^2(y^2 - 16) + 4(y^2 - 16) \\ &= (y^2 - 16)(y^2 + 4) \\ &= [(y)^2 - (4)^2](y^2 + 4) \\ &= \boxed{(y-4)(y+4)(y^2+4)} \text{ Ans} \end{aligned}$$